Software Requirements Specification (SRS)

PlateUp: AI-Powered Surplus Food Marketplace

Course: Web Programming Language
Project Type: Web-Based Application
Version: 1.0
Technology Stack: Node.js + Express.js + React.js + PostgreSQL + Python/Scikit-learn + Tailwind CSS

1. Introduction

1.1 Purpose

PlateUp is a web-based platform designed to reduce food waste by connecting restaurants, cafés, bakeries, and other food businesses with customers interested in purchasing surplus food at discounted prices. Businesses can list surplus food, while customers can discover and reserve available items. Machine learning will support demand prediction, surplus estimation, pricing recommendations, and personalized food recommendations.

1.2 Problem Statement

Food businesses frequently prepare more food than they can sell, resulting in avoidable food waste and financial loss. Customers may simultaneously be interested in quality food at lower prices. PlateUp addresses this gap through a dedicated surplus-food marketplace supported by AI-driven decision making.

1.3 Objectives

1. Develop a web-based marketplace connecting food businesses with customers seeking discounted surplus food.
2. Reduce avoidable food waste by facilitating timely sale of surplus food.
3. Help businesses recover revenue from unsold food.
4. Use machine learning to predict food demand and potential surplus.
5. Provide AI-assisted pricing recommendations.
6. Provide personalized surplus-food recommendations.

2. Scope

2.1 In Scope

Customer and business registration/login; business profile management; surplus food listing and management; food search, filtering, and discovery; location-based discovery; reservation/order management; reviews and ratings; business analytics; AI-based demand/surplus prediction; AI-based pricing recommendations; personalized recommendations; administrator management and monitoring.

2.2 Out of Scope

Dedicated Android/iOS applications, food delivery and driver tracking, advanced online payment systems, automated inventory hardware, AI chatbot, and automated food-quality inspection. These may be considered in future versions.

3. Overall System Description

3.1 System Description

PlateUp will consist of a React.js frontend, Node.js/Express.js backend, PostgreSQL database, and Python-based machine learning components.

3.2 User Roles

Customer: Searches for, reserves, and reviews surplus food.
Food Business: Publishes surplus food, manages listings/orders, and receives AI insights.
Administrator: Manages users, businesses, listings, orders, and platform activity.

3.3 Architecture

React.js Frontend → REST API (Node.js + Express.js) → PostgreSQL, with Python/Scikit-learn ML components providing demand prediction, dynamic pricing, and personalized recommendations.

4. Functional Requirements

4.1 Authentication and User Management

FR-01: The system shall allow customers and food businesses to register accounts.
FR-02: The system shall allow registered users to log in securely.
FR-03: The system shall provide role-based access for customers, businesses, and administrators.
FR-04: Users shall be able to view and update profile information.

4.2 Customer Functions

FR-05: Customers shall be able to browse available surplus food listings.
FR-06: Customers shall be able to search and filter listings by category, price, availability, business, and location.
FR-07: Customers shall be able to view food details including original price, discounted price, quantity, availability period, and pickup location.
FR-08: Customers shall be able to reserve available surplus food.
FR-09: The system shall update available quantity after successful reservation.
FR-10: Customers shall be able to view reservation/order history.
FR-11: Customers shall be able to rate and review businesses after completed transactions.

4.3 Food Business Functions

FR-12: Businesses shall be able to create and manage business profiles.
FR-13: Businesses shall be able to create surplus food listings containing food name, description, category, quantity, original price, discounted price, and availability period.
FR-14: Businesses shall be able to update, deactivate, or remove listings.
FR-15: Businesses shall be able to view and manage customer reservations.
FR-16: Businesses shall be able to view basic sales and surplus analytics.

4.4 AI/ML Functions

FR-17: The system shall analyze historical sales information to predict expected demand.
FR-18: The system shall estimate potential surplus from predicted demand and available/prepared quantity.
FR-19: The system shall generate recommended discounted prices using factors such as original price, predicted demand, remaining quantity, and remaining availability time.
FR-20: Businesses shall be able to review and modify AI-generated price recommendations before publishing.
FR-21: The system shall recommend relevant surplus food based on previous orders, preferences, price range, and location.

4.5 Administrator Functions

FR-22: Administrators shall be able to manage customer and business accounts.
FR-23: Administrators shall be able to monitor and manage food listings and reservations.
FR-24: Administrators shall be able to view basic platform statistics.

5. Non-Functional Requirements

5.1 Performance

NFR-01: The system should provide responsive performance for common operations. NFR-02: The system should support multiple concurrent users within the expected project scale.

5.2 Security

NFR-03: User passwords shall be securely hashed. NFR-04: Protected functions shall require authentication and authorization. NFR-05: Users shall only access resources permitted by their role.

5.3 Usability

NFR-06: The system shall provide an intuitive interface. NFR-07: The web application shall be responsive across desktop, tablet, and mobile browsers.

5.4 Reliability

NFR-08: The system shall maintain consistent inventory quantities during reservations. NFR-09: The system shall prevent reservations of unavailable quantities.

5.5 Maintainability & Scalability

NFR-10: The application shall use a modular architecture. NFR-11: Source code shall follow consistent coding/documentation practices. NFR-12: The architecture should allow additional customers and businesses without major architectural changes.

6. Data Requirements

7. Main Use Cases

8. Technology Requirements

9. Constraints and Assumptions

10. Future Enhancements

11. Expected Outcome

6. Data Requirements

Entity	Main Information
User	ID, name, email, password, role, location
Business	ID, name, address, contact, location, operating hours
Food Listing	Food, category, quantity, original price, rescue price, availability
Order	Customer, listing, quantity, status, timestamp
Review	Customer, business, rating, comment
Sales Data	Item, quantity sold, price, date/time
ML Results	Demand prediction, surplus estimate, price recommendation

7. Main Use Cases

UC-01: Customer Purchases Surplus Food. Customer logs in → searches/browses food → selects listing → views details → reserves quantity → system verifies availability → reservation is confirmed and inventory updated.

UC-02: Business Publishes Surplus Food. Business logs in → creates listing → enters quantity, price, and availability → system validates → business publishes listing.

UC-03: AI Demand Prediction. System retrieves historical sales → ML model analyzes features → predicts demand → estimates surplus → displays prediction to business.

UC-04: AI Pricing Recommendation. System obtains predicted demand and listing information → AI component recommends discount → business reviews/edits → listing is published.

8. Technology Requirements

Component	Technology
Frontend	React.js
UI Styling	Tailwind CSS
Backend	Node.js
API	Express.js
Database	PostgreSQL
Machine Learning	Python + Scikit-learn
Version Control	Git + GitHub

9. Constraints and Assumptions

Constraints: The initial system is web-only; it focuses primarily on pickup-based transactions; AI functionality is limited to features achievable within the academic timeline; prediction quality depends on sales-data quality.
Assumptions: Businesses provide accurate listing/sales information; customers collect reserved food within the specified period; businesses are responsible for food quality and safety; historical or synthetic sales data will be available for initial ML development.

10. Future Enhancements

Dedicated mobile applications; online payment integration; food delivery and real-time tracking; food donation to charities; advanced demand forecasting; loyalty/reward programs; geographic surplus-food heatmaps; advanced business intelligence; restaurant inventory integration.

11. Expected Outcome

The final PlateUp system is expected to provide a functional web-based marketplace where food businesses can list surplus food and customers can discover and reserve discounted food. Machine learning will differentiate the platform from a conventional food marketplace through demand prediction, surplus estimation, dynamic pricing recommendations, and personalized food recommendations.